

Payment Intelligence.

Dusk watches the financial activity flowing through a financial organization — the payments, transfers and fund movements that actually occur — and renders a verdict on each one. It enriches every actor with context from HR systems and the vendor / payee master, and it captures the growing **automation layer**: the finance copilots and payment bots that move or approve funds on someone's behalf.

TIME BUDGET

1–2 days · self-paced

STAGE

Post first interview

DATA

Synthetic — provided below

DELIVERABLESHi-fi Figma · Presentation ·
Dev-ready artifact**SYSTEMS DUSK DRAWS FROM**

ERP & ledgers (NetSuite, SAP, Workday) · payment rails (ACH, SWIFT, wire, card networks) · treasury & AP platforms — enriched with identity context from HR and the vendor / payee master, and the automation view from finance copilots and payment bots.

00 BEFORE YOU START

What this is, and how we'll read it

A NOTE ON SCOPE

There are three challenges below. If you only have time to make one truly excellent and sketch the other two, do that. We'd rather see one screen designed with conviction than three designed halfway.

What we're testing

- ◇ **Information design** — making complexity legible and scannable.
- ◇ **Product judgment** — what to surface, what to defer, what to cut.
- ◇ **Interaction & flow** — from noticing a risk to resolving it.
- ◇ **Craft** — hierarchy, states, and the details that build trust.
- ◇ **Communication** — can you defend a decision with a reason.

Ground rules

- ◇ Depth of thinking beats surface area — a strong slice beats a shallow whole.
- ◇ Use the synthetic dataset provided here.
- ◇ AI tools are welcome — we use them too. Just tell us where and how you used them.
- ◇ You don't need working integrations or real logos. Fidelity of thinking is the point.
- ◇ If something's ambiguous, make a call and state your assumption. That's part of the job.

01 THE PROBLEM

Valid credentials, used in ways they shouldn't be

Most fraud tooling asks *can this transaction clear* — is it authorized, is the balance there, does it pass the rules. Dusk asks a harder question that comes after the money has already moved: **which actor actually moved which funds — and should they have?**

Every financial event carries one verdict — a confidence, and the reasons behind it: ● Fraudulent ● Suspicious ● Legitimate

02 THE RAW MATERIAL

What the platform already knows

A Account & fund intelligence

FIELD	WHAT IT TELLS YOU	EXAMPLE
<code>summary</code>	One or two lines describing what the account actually is — generated, not guessed from the name.	<i>"FY24 executive bonus & equity payroll disbursement account."</i>
<code>topics[]</code>	Entities and themes detected inside — people, projects, cost centers, subjects.	<i>compensation · equity · treasury · payroll</i>
<code>ledger_class</code>	A universal chart-of-accounts code so unrelated accounts become comparable by function.	<i>641 — Payroll & compensation</i>
<code>value_tier</code>	Exposure classification: Routine · Elevated · High-Value · Critical.	<i>Critical, Routine, High-Value, Elevated</i>
<code>data_classes[]</code>	Regulated / high-value data types tied to the account.	<i>PII · PCI · financial · HR</i>
<code>compliance[]</code>	Regimes the activity falls under.	<i>SOX · BSA/AML · PCI-DSS · GDPR</i>
<code>owner / location</code>	Where it lives and who owns it in the source system.	<i>C. Rivas · NetSuite / Treasury</i>
<code>exposure</code>	How widely funds can leave: internal only, linked payees, external beneficiaries, open payout.	<i>2 external beneficiaries</i>
<code>access_population</code>	How many — and which — actors can move funds through it.	<i>41 actors · 3 external, 100 automated</i>
<code>risk_score</code>	Composite of value × exposure × transaction legitimacy.	<i>High · 87</i>

DESIGN TENSION WORTH NOTICING

An account is high-value, widely exposed, and being transacted against unusually all at once. Value, exposure and transaction legitimacy are three different axes — a good design lets someone reason across them without drowning.

B Actor intelligence

- ◇ **Type & context:** Employee (role, department, manager, tenure, status: active / notice / offboarded) · External / partner (vendor or personal payee accounts) · Service / automated (settlement & reconciliation jobs, NHI) · AI agent (autonomous actors moving or approving funds on someone's behalf).
- ◇ **Baseline & signals:** normal pattern (hours, geography, device, counterparties & transaction types) · entitlement breadth (what it could move vs. what it moves) · risk indicators (dormant-then-active, off-baseline counterparty, volume spikes) · actor risk score.

C The financial event

FIELD	DETAIL
actor → resource	Which actor moved funds through or against which account.
action	view · initiate · approve · transfer · route · withdraw · export — with volume and sequence.
context	Timestamp, app / rail (ACH · wire · card · SWIFT), source IP, geo, device.
verdict	
reasons[]	The why behind the verdict — e.g. “external beneficiary,” “off-baseline counterparty,” “during notice period,” “12× normal volume.” This is what makes a verdict trustworthy instead of a black box.

D Ask anything — the natural-language layer

- ◇ **Global** — ask from anywhere; a question that reaches across the whole environment. *“Which external beneficiaries received High-Value transfers last week?”*
- ◇ **In-context** — ask about what you're looking at, scoped to the record in view. *“Has anyone outside Treasury moved funds through this account before?”*

THE INTERESTING QUESTION

Natural language is powerful but non-deterministic; screens are trustworthy but rigid. Where does “ask” belong, and where must a fixed screen win? How does an answer hand back to the structured views — and how do you make an AI-generated answer auditable rather than something a fraud team has to take on faith? We want your point of view.

Three connected surfaces, one story

Understand → **prioritize** → **act**. Design them so they feel like one product. You choose the fidelity balance, but at least one screen should be genuinely hi-fi.

CHALLENGE 1

The payment-intelligence dashboard

Goal: understand → prioritize → act. The home surface where the Head of Fraud and the finance-security team live.

Design the primary experience where someone lands, grasps their exposure, and decides what to do first. Two very different people use it: a **Head of Fraud** who checks in weekly and needs the posture in 30 seconds, and a **fraud analyst** who works in it daily and needs to dig. You don't have to design both fully — but show us you've thought about the difference.

1. **Flow Intelligence** — Which actor groups are moving funds to and through which account categories — with context on what was touched, and drill-down from a group (actor / account) down to the individual actor, account, and event.
2. **Findings** — The events judged Fraudulent or Suspicious — the queue that needs a human. (Finding examples: Insider Payment Fraud · Account Takeover · Shadow Automation · Mandate Drift · Benign Anomaly.)
3. **Value-tier exposure** — Financial activity across the whole organization (any actor) broken out by tier: Routine · Elevated · High-Value · Critical, with the ability to drill down into the relevant items related to this potential level of exposure
4. **Automation adoption vs. fraud exposure** — Growth in AI-agent and automated payment initiation plotted against exposure over time — the widening blind spot.
5. **Most-exposed critical accounts** — The high-value accounts with the broadest access and the most diverse set of actors transacting against them — where a single compromise hurts most.

- ◇ Make it clear where the risk is and what to do next.
- ◇ Keep a dense inventory of actors, accounts, and events scannable — not a wall of rows.
- ◇ Express the legitimacy verdict visually so fraudulent activity is impossible to miss.
- ◇ Place the “Further investigate” query surface into the experience — how it's invoked, and how an answer returns the user to the structured views.
- ◇ Differentiate between further investigation through the prompt and the investigation flow map.

CHALLENGE 2

Taking action on a finding

Goal: seeing a problem is half the job. Design the other half.

Safe / reversible · Mark legitimate · Add to watchlist.

Workflow · Open case (Jira / ServiceNow) · Notify (Slack) · Push to fraud-ops queue.

Destructive / high-stakes · Freeze account · Revoke approval authority / right-size entitlement · Hold or reverse a payment · Block external beneficiary · Kill session.

- ◇ A clear model for single vs. bulk action, and for safe vs. destructive.
- ◇ What confirmation, preview-of-impact, or undo does a high-stakes action need?
- ◇ Show the **after** — the state once the action is taken. What changed, and how does the user know it worked?
- ◇ Further investigation may lead from any finding to a prompt-like surface where the context is already given and questions can be specifically asked.

CHALLENGE 3

The investigation: how did this happen?

Goal: reconstruct the chain of events behind a finding, and act along the way.

When a finding is serious, “what happened” isn’t enough — a responder needs to see *how it came to happen*: the approval authority that was granted, the account that went dormant then woke up, the accounts it reached, where the funds live and where they moved, the external transfer, the withdrawal. Design an investigation experience — a flow, a graph, a timeline, or your own invention — that reconstructs that story and lets the user take action at any point in it.

- ◇ Show the connections between actors, accounts, and events — not just a flat list.
- ◇ Make the sequence and the pivotal moment legible.
- ◇ Show where the funds are — their location in the source system, and how they moved (transferred, routed, withdrawn, taken outside the org).
- ◇ Let the responder act mid-investigation — freeze the actor, hold a payment, kill a session — without losing the thread.
- ◇ Leave a defensible record: what was found, what was done, why.

04 A SCENARIO TO DESIGN AGAINST

Vega Dynamics — a seeded incident

Vega Dynamics

Synthetic mid-size company · ~3,200 employees · running Dusk across its payment & treasury systems

THE SITUATION

A finance analyst, **D. Marek**, is 9 days into a 30-day notice period. Six months ago a broad approval authority over the Finance disbursement accounts was granted for a quarter-close project and never revoked. This week that authority woke up: a burst of access across Critical compensation and payroll accounts — data Marek’s baseline never touched — followed by two transfers routed to a personal account not in the payee master, then confirmed and withdrawn from an unrecognized device in a new location. Dusk flagged the chain • **Fraudulent** • **0.91**.

-6 months **Approval authority granted** — broad disbursement rights over Finance accounts added for quarter-close. Never revoked. Sits dormant.

Day 1 of 30 **Notice period begins** — actor status flips to notice. Baseline: rarely touches comp / payroll accounts.

Tue 09:12 **Dormant authority reactivates** — 14 Critical finance accounts accessed and payments queued in 20 minutes (12× normal volume, off-baseline counterparty).

Tue 09:40 **External transfer created** — two payments routed to a personal account not in the payee master.

Tue 22:07 **Funds withdrawn** — the external transfer is confirmed and drawn down from an unrecognized device in a new geography.

05 APPENDIX

Sample data to design with

Accounts

ACCOUNT	CLASS	TIER	DATA	EXPOSURE	RISK
FY24 Comp & Equity Payroll	641	Critical	PII · financial · HR	2 external	87
Payroll Master — EU	642	Critical	PII · financial	internal	72
Board Reserve — Q3	658	High-Value	financial · strategy	6 actors	61
Settlement API Credentials	000	Critical	secrets / keys	exposed key	93
Vendor Payments 2024	646	High-Value	PII · legal	3 external	55
Team Offsite Fund	650	Elevated	—	internal	22
Petty Cash — Marketing	659	Routine	—	open payout	6

Actors

ACTOR	TYPE	CONTEXT	SIGNAL	RISK
D. Marek	employee	Finance · on notice	dormant → active	88
j.doe.personal@gmail	external	not in payee master	restricted transfer	81
svc-settle-03	service / automated	nightly settlement job	scope drift	64
agent-treasury	AI agent	treasury copilot	off-baseline counterparty	58
C. Rivas	employee	Finance · owner	in baseline	12

Financial events

ACTOR → RESOURCE	ACTION	WHEN	VERDICT	WHY
D. Marek → Comp & Equity Payroll	access ×14 / queue	Tue 09:12	● Fraudulent	notice · off-baseline · 12× vol
D. Marek → Payroll Master	transfer → personal	Tue 09:40	● Fraudulent	external target · restricted
j.doe.personal → Payroll Master	withdraw	Tue 22:07	● Fraudulent	new device · new geo
agent-treasury → Board Reserve Q3	view / move	Wed 03:20	● Suspicious	off-hours · off-baseline
C. Rivas → Comp & Equity Payroll	approve	Mon 14:02	● Legitimate	owner · in baseline

06 THE BAR

What a great submission looks like

- ◆ **Clarity under density** — The hardest, most crowded moment still reads at a glance. Risk and next-action are never buried.
- ◆ **Decisions, with reasons** — You can say why the verdict is here and not there, why this action is one click and that one is guarded.
- ◆ **A coherent system** — The three surfaces feel like one product. A pattern learned once carries across.
- ◆ **Trust by design** — Verdicts are explainable, AI answers are auditable, destructive actions are hard to do by accident.
- ◆ **Craft in the details** — States, empty states, loading, the “after” of an action — the parts that separate a mock from a product.
- ◆ **A point of view** — You made real calls, took a considered risk somewhere, and can defend it. We want to meet a designer, not a template.

07 WHAT TO SUBMIT

Deliverables & how to send them

- 1 **Hi-fi Figma** — Your screens for the challenges, at least one taken to genuine high fidelity. Lower-fidelity frames or flow sketches for the rest are welcome. Leave your explorations visible; we like seeing the thinking.
- 2 **Presentation** — A short deck that walks us through your reasoning: the problem as you framed it, the key decisions and their benefit, the trade-offs you made, and what you’d do next with more time.
- 3 **Dev-ready artifact** — A structured, data-driven version of a key screen — a coded prototype, a component built against a data schema, or a clearly-specified data-to-UI mapping. Show that your design is built to be filled with live data. You should also demonstrate it with an HTML artifact

ON THE DEV-READY ARTIFACT

We're not asking for production code. We want to see what you think in terms of **data** → **interface**: what fields drive each element, what an empty or overflowing state does, how a developer would wire the sample data (from the Appendix) into your design without guessing. Use whatever medium shows that best.